

The Civic Record Evaluation Framework (CREF) Version 2.0

A Deterministic, Explainable, and Reflexively Self-Describing Framework for Structured Evaluation of Visible Civic Records

Abstract

This paper presents Version 2.0 of the Civic Record Evaluation Framework (CREF), a deterministic framework for the structured evaluation of visible civic records and existing record-derived outputs.

Version 1.0 introduced CREF as a methodology for evaluating visible record structures through deterministic classifications, administrative pathways, traceability mechanisms, and reproducible outputs. Version 2.0 extends this foundation by introducing methodological provenance, evidence attribution, explainability certification, and framework self-description capabilities.

The framework now exposes determination pathways, identifies supporting evidence for existing outputs, provides visible rule citations, certifies explainability component availability, and deterministically describes its own implemented methodology.

Importantly, these capabilities do not introduce new reasoning, modify framework behaviour, infer facts, determine truth, assign liability, validate evidence, or perform self-improvement. All outputs remain entirely constrained by visible inputs, documented rules, implemented stage logic, and declared methodological boundaries.

Version 2.0 therefore presents a deterministic framework that not only produces record-derived outputs but can also expose, inspect, and describe the methodology by which those outputs are generated. The result is a framework that is simultaneously deterministic, inspectable, reproducible, explainable, and reflexively self-describing.

1. Introduction

The evaluation of civic records frequently requires methods that provide transparency, traceability, reproducibility, and explainability without relying upon undocumented assumptions or external truth determinations.

Many computational systems generate outputs while providing limited visibility into the pathways by which those outputs were produced. As a result, inspection of methodology, evidence support, and procedural boundaries may remain incomplete.

The Civic Record Evaluation Framework (CREF) was developed to address this problem.

Version 1.0 introduced a deterministic methodology for the structured evaluation of visible civic record data.

Version 2.0 extends this methodology by enabling the framework to expose and describe its own implemented architecture while remaining entirely constrained by documented methodology and declared boundaries.

The objective is not artificial reasoning or autonomous interpretation.

The objective is deterministic methodological transparency.

2. Framework Principles

CREF Version 2.0 operates according to five fundamental principles.

Determinism

The same visible inputs follow the same documented rules and produce the same outputs.

Traceability

Visible evidence, governance decisions, and record evolution pathways remain inspectable.

Explainability

Determination pathways and supporting components remain exposed.

Reproducibility

Analysis may be repeated from the same visible inputs.

Boundary Transparency

Framework capabilities and limitations remain explicitly declared.

3. Framework Scope

Inputs

The framework accepts five declared input categories:

- Visible record data
- Metadata
- Relationships
- Conditions
- Existing framework outputs

These inputs are limited to visible record-derived information and previously implemented framework outputs.

Outputs

The framework produces nine declared output categories:

- Conditions
- Trajectories
- Administrative outputs
- Record evolution classifications
- Determination traces
- Rule citations
- Evidence attribution views
- Explainability outputs
- Certification outputs

All outputs are derived through deterministic processing only.

4. Framework Architecture

Version 2.0 consists of eleven implemented layers.

Evaluation Layers

1. Visible Record Layer

Establishes visible record identifiers, metadata, and evidence context.

2. Conditions Layer

Produces deterministic condition outputs from visible inputs.

3. Pattern Interpretation Layer

Produces pattern interpretation outputs from visible signals and relationships.

4. Trajectory Classification Layer

Produces trajectory and system-state classifications.

5. Administrative Evaluation Layer

Produces administrative actions, workflow states, review pathways, and archive outputs.

6. Record Evolution Layer

Produces visible record lineage and evolution-chain classifications.

Reflexive Layers

7. Determination Trace Layer

Exposes the ordered pathway from visible record inputs to existing determinations.

8. Rule Citation Layer

Exposes visible deterministic rule families and citations supporting determination pathways.

9. Evidence Attribution Layer

Exposes visible evidence-to-output attribution and unsupported output visibility.

10. Explainability Certification Layer

Certifies the availability of explainability components and methodological provenance outputs.

11. Framework Self-Description Layer

Produces deterministic descriptions of implemented framework methodology.

Together, these eleven layers transform visible record inputs into structured, inspectable, record-derived outputs through deterministic processing only.

5. Methodological Provenance

Version 2.0 introduces methodological provenance mechanisms that expose:

- Determination pathways
- Visible evidence attribution
- Rule family citations
- Explainability component availability
- Framework guarantees
- Methodological limitations
- Framework self-description outputs

The framework therefore becomes capable of exposing not only its outputs but also the implemented methodology responsible for those outputs.

6. Reflexive Methodology

Version 2.0 introduces deterministic framework self-description.

The framework can describe:

- Its identity
- Its purpose
- Its scope
- Its implemented architecture
- Its guarantees
- Its constraints
- The methodology by which outputs are derived

This description is generated through deterministic processing of documented methodology and implemented stage definitions.

No new reasoning is introduced.

No self-modification occurs.

No self-improvement occurs.

The framework describes only what already exists within the implemented framework.

7. Methodological Boundaries

Version 2.0 does not:

- Determine truth
- Determine liability
- Infer intent
- Assign blame
- Validate evidence
- Determine factual correctness
- Determine legal sufficiency
- Determine real-world sufficiency
- Determine real-world evidential sufficiency
- Create evidence
- Create rules
- Create conditions
- Create classifications
- Create methodology
- Create framework identity
- Introduce new reasoning
- Perform self-improvement
- Change framework behaviour

All outputs remain constrained by documented methodology and visible record-derived inputs.

8. Reflexive Consistency

A central contribution of Version 2.0 is reflexive consistency.

Version 1.0 established CREF as a deterministic methodology for the structured evaluation of visible civic records. It defined the framework's principles, inputs, outputs, evaluation layers, guarantees, and operational boundaries.

Version 2.0 demonstrates that the same deterministic principles governing record evaluation can also be applied to the framework itself.

The eleven-layer architecture makes this extension explicit.

Layers 1–6 evaluate visible records and produce structured record-derived outputs.

Layers 7–11 expose and describe the methodology responsible for producing those outputs.

The distinction between these two groups is fundamental. The framework becomes capable of inspecting and describing itself through the same deterministic principles that govern its evaluation of records.

This capability does not introduce new reasoning, modify framework behaviour, generate methodology, or perform self-improvement. The reflexive layers remain entirely constrained by documented rules, visible inputs, implemented stage logic, and declared methodological boundaries.

The result is reflexive consistency: a framework that remains deterministic and reproducible not only when applied to records but also when applied to its own methodology.

The framework's claims concerning determinism, transparency, explainability, and reproducibility are therefore not merely asserted. They are demonstrated by the framework's own behaviour.

Version 2.0 consequently represents a transition from a deterministic evaluation framework to a deterministic, explainable, and reflexively self-describing methodology whose principles remain internally consistent when recursively applied to itself.

9. Contributions of Version 2.0

Version 2.0 contributes:

1. Deterministic determination traces.
2. Rule citation visibility.
3. Evidence attribution matrices.
4. Explainability certification.
5. Framework self-description.
6. Reflexive methodological transparency.
7. Explicit methodological boundaries.
8. Fully inspectable record-derived outputs.
9. Demonstrated reflexive consistency.

Collectively, these capabilities transform CREF from a deterministic evaluation framework into a deterministic, explainable, and reflexively self-describing methodology.

10. Conclusion

Version 1.0 introduced a deterministic framework for the structured evaluation of visible civic records.

Version 2.0 demonstrates that the framework can expose, inspect, and describe its own implemented methodology while remaining entirely constrained by documented rules, visible inputs, and declared boundaries.

The framework therefore does not merely produce outputs.

It can expose the pathways that produced those outputs.

It can attribute visible evidence supporting those outputs.

It can identify the rule families governing those outputs.

It can certify the availability of explainability components.

And it can describe the methodology that makes those capabilities possible.

The framework's claims about determinism, transparency, explainability, and reproducibility are therefore not simply declared. They are demonstrated by the framework's own deterministic behaviour when describing itself.

Version 2.0 therefore presents a deterministic framework that can expose, inspect, and describe its own implemented methodology without introducing new reasoning.